WORKED SOLUTION

STEP I 1988, Question 12

Mechanics • Variable Force Motion

STEP I 1988, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q12

The key idea

Work out, for each method, the furthest the skater travels towards the bank; she is safe exactly when d exceeds that distance for at least one method. Turning is classic circular motion: a constant force perpendicular to the velocity gives a circle of fixed radius, and the radius is the distance she still advances. Braking is a variable-force problem in which the force depends on speed but we want a *distance*, which is the cue to write the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}x}$.

Solution

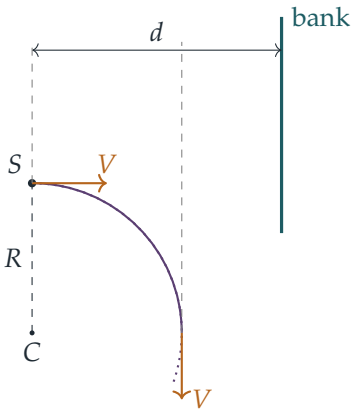
The model

We treat the skater as a particle of mass M on smooth horizontal ice. The only horizontal force on her is the one she applies, so weight and the normal reaction cancel and play no further part. We measure distance x from her starting point towards the bank, so the bank is at $x = d$.

Method 1: turning in a circle

A force perpendicular to the velocity does no work, so her speed stays V. A force of constant magnitude F always at right angles to a velocity of constant speed V provides the centripetal force for a circle of radius R, where

$$F = \frac{MV^2}{R}, \qquad \text{so} \qquad R = \frac{MV^2}{F}.$$



She starts at S moving straight at the bank, so the centre C of the circle lies on the line through S *parallel* to the bank, at distance R from S. Her distance towards the bank is then $x = R\sin\theta$ after turning through angle θ, which is greatest, equal to R, when $\theta = \frac{\pi}{2}$; there she is moving parallel to the bank, and afterwards she curves away. So she misses the bank if

$$d > \frac{MV^2}{F}.$$

(If $d = MV^2/F$ exactly she just grazes the bank, moving along it.)

Method 2: braking

Now the force acts directly backwards along her line of motion, so she moves in a straight line towards the bank and slows down. Newton's second law, with the acceleration written as $v\frac{\mathrm{d}v}{\mathrm{d}x}$ because we want distance rather than time, gives

$$Mv\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{F(V^2+v^2)}{2V^2}.$$

Separating the variables and integrating from $x = 0$, where $v = V$, to the point $x = X$ where she stops:

$$\int_V^0 \frac{v}{V^2+v^2}\,\mathrm{d}v = -\frac{F}{2MV^2}\int_0^X \mathrm{d}x,$$
$$\left[\tfrac{1}{2}\ln(V^2+v^2)\right]_V^0 = -\frac{FX}{2MV^2},$$
$$\tfrac{1}{2}\ln\frac{V^2}{2V^2} = -\tfrac{1}{2}\ln 2 = -\frac{FX}{2MV^2},$$

so her stopping distance is

$$X = \frac{MV^2}{F}\ln 2.$$

Ask yourself Does she actually come to rest, or does she only approach zero speed as she creeps forward for ever?

The braking force never drops below $\frac{1}{2}F$, so her speed falls to zero in a finite time. Explicitly, $M\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{F}{2V^2}(V^2+v^2)$ integrates to $\frac{2MV}{F}(\arctan 1 - \arctan\frac{v}{V}) = t$, so she stops at $t = \pi MV/(2F)$, having gone the finite distance X. (Once at rest she stops pushing.) So braking saves her when $d > X$; if $d = X$ she reaches the bank with zero speed, touching it without any impact.

Comparing the methods

Since $\ln 2 \approx 0.693 < 1$, the stopping distance is shorter than the turning radius: braking is the better method, and whenever turning works, braking works too. Hence she can avoid hitting the bank precisely when

$$\boxed{d > \frac{MV^2}{F}\ln 2}$$

(with $d = \frac{MV^2}{F}\ln 2$ the borderline case in which she arrives at the bank at rest).

The particle assumption

A real skater has size: her body and arms reach some way in front of her centre of mass, so in practice she needs d larger than the value above by roughly that reach. More seriously, the forces come from her skates on the ice. To turn she must lean into the curve, and to brake hard she must twist her blades across her motion (a hockey stop), which turns her body sideways. Rotation and posture like this are exactly what a particle model ignores. The model also quietly assumes the ice can supply these forces at all, although it is called smooth; the blades grip sideways even when there is no friction along them.

Checks

- Units: MV^2/F is $\mathrm{kg\,m^2\,s^{-2}/N} = \mathrm{m}$, a length, as it must be.
- Bounds: the braking force lies between $\frac{1}{2}F$ and F. A constant force F would stop her in $MV^2/(2F)$ and a constant $\frac{1}{2}F$ in MV^2/F, and indeed $\frac{1}{2} < \ln 2 < 1$.
- Numerically: stepping both motions with a Runge–Kutta integrator for random M, V, F (for example $R = MV^2/F = 7.0747$) gave furthest advance 7.0747 when turning and stopping distance $4.9038 = R\ln 2$ when braking, with stopping time $\pi MV/(2F)$.

Common slips

- Placing the centre of the circle on the line of motion (or at the bank), and getting a reach of $2R$ or 0, rather than putting the centre beside her so that the reach is R.
- Using $\frac{\mathrm{d}v}{\mathrm{d}t}$ for the braking and then struggling to get a distance; $v\,\frac{\mathrm{d}v}{\mathrm{d}x}$ gives it in one integration.
- Giving two separate conditions without comparing them. The question asks for the set of d for which she *can* escape, so we need the better of the two methods.

Beyond STEP

Writing $\ddot{x} = v\,\frac{\mathrm{d}v}{\mathrm{d}x}$ is the first step towards energy methods: for a force depending only on position it is the work–energy theorem, and in university mechanics it develops into potentials and phase-plane pictures of motion. The skater's choice between steering and braking is also a simple optimal control problem: what is the best way to use a limited force to avoid an obstacle? Kleppner and Kolenkow, *An Introduction to Mechanics*, treats velocity-dependent forces and circular motion at exactly the next level up.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.